19.2: Viruses replicate only in host cells

Viruses lack metabolic enzymes and equipment for making proteins, such as ribosomes. They are obligate intracellular parasites; in other words, they can replicate only within a host cell. It is fair to say that viruses in isolation are merely packaged sets of genes in transit from one host cell to another.

Each particular virus can infect cells of only a limited number of host species, called the **host range** of the virus. This host specificity results from the evolution of recognition systems by the virus. Viruses usually identify host cells by a “handshake” fit between viral surface proteins and specific receptor molecules on the outside of cells. Such receptor molecules are most often proteins that carry out necessary functions for the host cell and have been adopted by viruses as portals of entry. Some viruses have broad host ranges. For example, West Nile virus and equine encephalitis virus are distinctly different viruses that can each infect mosquitoes, birds, horses, and humans. Other viruses have host ranges so narrow that they infect only a single species. Measles virus, for instance, can infect only humans. Furthermore, viral infection of multicellular eukaryotes is usually limited to particular tissues. Human cold viruses infect only the cells lining the upper respiratory tract, and the HIV seen in Figure 19.1 binds to receptors present only on certain types of immune cells.

General Features of Viral Replicative Cycles

A viral infection begins when a virus binds to a host cell and the viral genome makes its way inside **(Figure 19.4)**. The mechanism of genome entry depends on the type of virus and the type of host cell. For example, T-even phages use their elaborate tail apparatus to inject DNA into a bacterium (see Figure 19.3d). Other viruses are taken up by endocytosis or, in the case of enveloped viruses, by fusion of the viral envelope with the host’s plasma membrane. Once the viral genome is inside, the proteins it encodes can commandeer the host, reprogramming the cell to copy the viral genome and manufacture viral proteins. The host provides the nucleotides for making viral nucleic acids, as well as enzymes, ribosomes, tRNAs, amino acids, ATP, and other components needed for making the viral proteins. Many DNA viruses use the DNA polymerases of the host cell to synthesize new genomes along the templates provided by the viral DNA. In contrast, to replicate their genomes, RNA viruses use virally encoded RNA polymerases that can use RNA as a template. (Uninfected cells generally make no enzymes for carrying out this process.)

Figure 19.4

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A simplified viral replicative cycle.
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Simplified viral replicative cycle; the parasite is a D N A virus with a capsid consisting of a single type of protein which enters the cell. New virus particles exit the cell.

A virus is an intracellular parasite that uses the equipment and small molecules of its host cell to replicate. In this simplest of viral cycles, the parasite is a DNA virus with a capsid consisting of a single type of protein.

Simplified viral replicative cycle; the parasite is a D N A virus with a capsid consisting of a single type of protein which enters the cell.

1. The virus enters the cell and is uncoated, releasing viral D N A and capsid proteins.
2. Host enzymes replicate the viral genome inside the cell.
3. Meanwhile, host enzymes inside the cell transcribe the viral genome into viral m R N A, which host ribosomes us to make more capsid proteins.
4. Viral genomes and capsid proteins self-assemble into new virus particles, which exit the cell.

MAKE CONNECTIONS *Label each of the straight gray arrows to identify the process that is occurring. Review**Figure 17.25.*

*For suggested answer, see*Appendix A.**

Animation

Simplified Viral Replicative Cycle

image

image

After the viral nucleic acid molecules and capsomeres are produced, they spontaneously self-assemble into new viruses. In fact, researchers can separate the RNA and capsomeres of TMV and then reassemble complete viruses simply by mixing the components together under the right conditions. The simplest type of viral replicative cycle ends with the exit of hundreds or thousands of viruses from the infected host cell, a process that often damages or destroys the cell. Such cellular damage and death, as well as the body's responses to this destruction, cause many of the symptoms associated with viral infections. The viral progeny that exit a cell have the potential to infect additional cells, spreading the viral infection.

There are many variations on the simplified viral replicative cycle we have just described. We will now take a look at some of these variations in bacterial viruses (phages) and animal viruses; later in the chapter, we will consider plant viruses.

Replicative Cycles of Phages

Phages are the best understood of all viruses, although some of them are also among the most complex. Research on phages led to the discovery that some double-stranded DNA viruses can replicate by two alternative mechanisms: the lytic cycle and the lysogenic cycle.

The Lytic Cycle

A phage replicative cycle that culminates in death of the host cell is known as a **lytic cycle**. The term *lytic* refers to the last stage of infection, during which the bacterium lyses (breaks open) and releases the phages that were produced within the cell. Each of these phages can then infect a healthy cell, and a few successive lytic cycles can destroy an entire bacterial population in just a few hours. A phage that replicates only by a lytic cycle is a **virulent phage**. **Figure 19.5** illustrates the major steps in the lytic cycle of T4, a typical virulent phage.

Figure 19.5

The lytic cycle of phage T4, a virulent phage.

The lytic cycle of phage T4: 1) Attachment. 2) Entry of phage D N A and degradation of host D N A. 3) Synthesis of viral genomes and proteins. 4) Self-assembly. 5) Release.

The lytic cycle of phage T4:

1. Attachment. The T4 phage uses its tail fibers to bind to specific surface proteins on an E. coli cell that act as receptors.
2. Entry of phase D N A and degradation of host D N A. The sheath of the tail contracts, injecting the phage D N A into the cell and leaving an empty capsid outside. The cell's D N A is hydrolyzed.
3. Synthesis of viral genomes and proteins. The phage D N A directs production of phage proteins and copies of the phage genome by host and viral enzymes, using components within the cell.
4. Self-assembly. Three separate sets of protein self-assemble to form phage heads, tails, and tail fibers. The phage genome is packaged inside the capsid as the head forms.
5. Release. The phage directs production of an enzyme that damages the bacterial wall, allowing fluid to enter. The cell swells and finally bursts, releasing 100 to 200 phage particles.

Phage T4 has almost 300 genes, which are transcribed and translated using the host cell’s machinery. One of the first phage genes translated after the viral DNA enters the host cell codes for an enzyme that degrades the host cell’s DNA (**STEP 2**); the phage DNA is protected from breakdown because it contains a modified

form of cytosine that is not recognized by the phage enzyme. The entire lytic cycle, from the phage's first contact with the cell surface to cell lysis, takes only 20–30 minutes at 37°C.

Animation

Phage Lytic Cycle

image

image

The Lysogenic Cycle

Instead of lysing their host cells, many phages coexist with them in a state called lysogeny. In contrast to the lytic cycle, which kills the host cell, the **lysogenic cycle** allows replication of the phage genome without destroying the host. Phages capable of using both modes of replicating within a bacterium are called **temperate phages**. A temperate phage called lambda, written with the Greek letter λ, has been widely used in biological research. Phage λ resembles T4, but its tail has only one short tail fiber.

Infection of an *E. coli* cell by phage λ begins when the phage binds to the surface of the cell and injects its linear DNA genome **(Figure 19.6)**. Within the host, the λ DNA molecule forms a circle. What happens next depends on the replicative mode: lytic cycle or lysogenic cycle. During a lytic cycle, the viral genes immediately turn the host cell into a λ-producing factory, and the cell soon lyses and releases its virus progeny. During a lysogenic cycle, however, the λ DNA molecule is incorporated into a specific site on the *E. coli* chromosome by viral proteins that break both circular DNA molecules and join them to each other. When integrated into the bacterial chromosome in this way, the viral DNA is known as a **prophage**. One prophage gene codes for a protein that prevents transcription of most of the other prophage genes. Thus, the phage genome is mostly silent within the bacterium. Every time the *E. coli* cell prepares to divide, it replicates the phage DNA along with its own chromosome such that each daughter cell inherits a prophage. A single infected cell can quickly give rise to a large population of bacteria carrying the virus in prophage form. This mechanism enables viruses to propagate without killing the host cells on which they depend.

Figure 19.6

The lytic and lysogenic cycles of phage λ, a temperate phage.

Lytic cycle and lysogenic cycles are detailed after the lambda D N A enters the bacterial cell.

After entering the bacterial cell the lambda D N A induce a lytic cycle or a lysogenic cycle.

Lytic cycle: New phage D N A and proteins are synthesized and self-assemble into phages. The cell lyses, releasing phages. The phage attaches to a host cell and injects its D N A through the tail fiber. The Phage D N

A circularizes.

Lysogenic cycle: Phage D N A integrates into the bacterial chromosome, becoming a prophage. The bacterium reproduces normally, copying the prophage and transmitting it to daughter cells. Many cell divisions produce a large population of bacteria infected with the prophage. Occasionally, a daughter cell with prophage exits the bacterial chromosome, initiating a lytic cycle.

After entering the bacterial cell and circularizing, the λ DNA can immediately initiate the production of a large number of progeny phages (lytic cycle) or integrate into the bacterial chromosome (lysogenic cycle). In most cases, phage λ follows the lytic pathway, which is similar to that detailed in Figure 19.5. However, once a lysogenic cycle begins, the prophage may be carried in the host cell's chromosome for many generations. Phage λ has one main tail fiber, which is short.

Animation

Phage Lysogenic and Lytic Cycles

image

image

The term *lysogenic* signifies that prophages are capable of generating active phages that lyse their host cells. This occurs when the λ genome (or that of another temperate phage) is induced to exit the bacterial chromosome and initiate a lytic cycle. An environmental signal, such as a certain chemical or high-energy radiation, usually triggers the switchover from the lysogenic to the lytic mode.

In addition to the gene for the viral protein that prevents transcription, a few other prophage genes may be expressed during lysogeny. Expression of these genes may alter the host's phenotype, a phenomenon that can have important medical significance. For example, the three species of bacteria that cause the human diseases diphtheria, botulism, and scarlet fever would not be so harmful to humans without certain prophage genes that cause the host bacteria to make toxins. And the difference between the *E. coli* strain in our intestines and the O157:H7 strain that has caused several deaths by food poisoning appears to be the presence of toxin genes of prophages in the O157:H7 strain.

Bacterial Defenses Against Phages

After reading about the lytic cycle, you may have wondered why phages haven't exterminated all bacteria. Lysogeny is one major reason why bacteria have been spared from extinction caused by phages. Bacteria also have their own defenses against phages. First, natural selection favors bacterial mutants with surface proteins that are no longer recognized as receptors by a particular type of phage. Second, when phage DNA does enter a bacterium, the DNA often is identified as foreign and cut up by cellular enzymes called

restriction enzymes, which are so named because they *restrict* a phage's ability to replicate within the bacterium. (Restriction enzymes are used in molecular biology and DNA cloning techniques; see Concept 20.1.) The bacterium's own DNA is methylated in a way that prevents attack by its own restriction enzymes. A third defense is a system present in both bacteria and archaea called the *CRISPR-Cas system*, which you learned about in Concept 17.5.

HHMI BioInteractive Video

Jennifer Doudna: CRISPR-Cas9 was First Discovered in Bacteria

image

image

HHMI BioInteractive Video

Jennifer Doudna: Science is a Serendipitous Endeavor

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The CRISPR-Cas system was discovered during a study of repetitive DNA sequences present in the genomes of many prokaryotes. These sequences, which puzzled scientists, were named clustered regularly interspaced short palindromic repeats (CRISPRs) because each sequence reads the same forward and backward (a palindrome), with different stretches of "spacer DNA" in between the repeats. At first, scientists assumed the spacer DNA sequences were random and meaningless, but analysis by several research groups showed that each spacer sequence corresponded to DNA from a particular phage that had infected the cell. Further studies revealed that particular nuclease proteins interact with the CRISPR region. These nucleases, called Cas (CRISPR-associated) proteins, can identify and cut phage DNA, thereby defending the bacterium against phage infection.

When a phage infects a bacterial cell that has the CRISPR-Cas system, the DNA of the invading phage is stored, integrated into the genome between two repeat sequences **(Figure 19.7)**. If the cell survives the infection, any further attempt by the same type of phage to infect this cell (or its offspring) triggers transcription of the CRISPR region into RNA molecules. These RNAs are cut into pieces and then bound by Cas proteins, such as the Cas9 protein (see Figure 17.28). The Cas protein uses a portion of the phage-related RNA as a homing device to identify the invading phage DNA and cut it, leading to its destruction.

Figure 19.7

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The CRISPR–Cas system: a type of bacterial immune system.
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Drawing summarizes the C R I S P R-C a s system in five steps.

The steps are as follows.

1.Infection by a phage triggers transcription of the C R I S P R region of the bacterial D N A. This region consists of D N A from phages that previously infected the cell, separated by repeats.

2. The R N A transcript is processed into short R N A strands. Here we focus on R N A that is complementary to the D N A of the invading phage.
3. Each short R N A strand binds to a Cas protein, forming a complex. (Multiple complexes are formed.)
4. Complementary R N A binds to D N A from the invading phage. The Cas protein then cuts the phage D N A.
5. After being cut, the entire phage D N A molecule is degraded and can no longer be replicated.

Interview

****Interview with Francisco Mojica: Discovering CRISPR in bacteria**

A photograph of Francisco Mojica, a Spanish molecular biologist.

Just as natural selection favors bacteria that have receptors altered by mutation or that have enzymes that cut phage DNA, it also favors phage mutants that can bind to altered receptors or that are resistant to enzymes. Thus, the bacterium-phage relationship is in constant evolutionary flux.

Replicative Cycles of Animal Viruses

Everyone has suffered from viral infections, whether cold sores, influenza, or the common cold. Like all viruses, those that cause illness in humans and other animals can replicate only inside host cells. Many variations on the basic scheme of viral infection and replication are represented among the animal viruses. Key variables are the nature of the viral genome (double- or single-stranded DNA or RNA) and the presence or absence of an envelope.

Whereas there are only a few bacteriophages that have an envelope or RNA genome, many animal viruses have both. In fact, nearly all animal viruses with RNA genomes have an envelope, as do some with DNA

genomes. Rather than consider all the mechanisms of viral infection and replication, we will focus first on the roles of viral envelopes and then on the functioning of RNA as the genetic material of many animal viruses.

Viral Envelopes

An animal virus equipped with an envelope—that is, a membranous outer layer—uses it to enter the host cell. Protruding from the outer surface of this envelope are viral glycoproteins that bind to specific receptor molecules on the surface of a host cell. For example, the Spike protein on SARS-CoV-2 binds to a protein called ACE2 on the surface of human airway (and other) cells. **Figure 19.8** outlines the events in the replicative cycle of an enveloped virus with an RNA genome. Ribosomes bound to the endoplasmic reticulum (ER) of the host cell make the protein parts of the envelope glycoproteins; cellular enzymes in the ER and Golgi apparatus then add the sugars. The resulting viral glycoproteins, embedded in membrane derived from the host cell, are transported to the cell surface. In a process much like exocytosis (see Figure 7.20), new viral capsids are wrapped in membrane as they bud from the cell. In other words, the viral envelope is usually derived from the host cell's plasma membrane, although all or most of the molecules of this membrane are specified by viral genes. The enveloped viruses are now free to infect other cells. This replicative cycle does not necessarily kill the host cell, in contrast to the lytic cycles of phages.

Figure 19.8

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The replicative cycle of an enveloped RNA virus.
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Replicative cycle of an enveloped R N A virus showing each of the stages and activities during the cycle.

Shown here is a virus with a single-stranded RNA genome that functions as a template for synthesis of mRNA.

Replicative cycle of an enveloped R N A virus: 1) Glycoproteins on the viral envelop bind to specific receptor molecules on the host cell, promoting viral uptake by the cell. 2) The capsid and viral genome enter the cell. Digestion of the capsid by cellular enzymes releases the viral genome (R N A). 3) The viral genome functions as a template for synthesis of complementary R N A strands by a viral R N A polymerase. 4) New copies of viral genome R N A are made using the complementary R N A strands as templates. 5) Complementary R N A strands also function as m R N A, which is translated into both capsid proteins (in the cytosol) and glycoproteins for the viral envelope (in the ER and Golgi apparatus). 6) Vesicles transport envelope glycoproteins to the plasma membrane. 7) A capsid assembles around each viral genome molecule. 8) Each

new virus buds from the cell, its envelope studded with viral glycoproteins embedded in membrane derived from the host cell.

Some viruses have envelopes that are not derived from plasma membrane. Herpesviruses, for example, are temporarily cloaked in membrane derived from the nuclear envelope of the host; they then shed this membrane in the cytoplasm and acquire a new envelope made from membrane of the Golgi apparatus. These viruses have a double-stranded DNA genome and replicate within the host cell nucleus, using a combination of viral and cellular enzymes to replicate and transcribe their DNA. In the case of herpesviruses, copies of the viral DNA can remain behind as mini-chromosomes in the nuclei of certain nerve cells. There they remain latent until some sort of physical or emotional stress triggers a new round of active virus production. The infection of other cells by these new viruses causes the blisters characteristic of herpes, such as cold sores or genital sores. Once someone acquires a herpesvirus infection, flare-ups may recur throughout the person's life.

Viral Genetic Material

Table 19.1 shows the common classification system for animal viruses, which is based on their genetic material: double- or single-stranded DNA, or double- or single-stranded RNA, with further classification of RNA viruses. Although some phages and most plant viruses are RNA viruses, the broadest variety of RNA genomes is found among the viruses that infect animals. There are three types of single-stranded RNA genomes found in animal viruses (classes IV–VI in Table 19.1). The genome of class IV viruses can directly serve as mRNA and thus can be translated into viral protein immediately after infection. (SARS-CoV-2 has this type of genome.) Figure 19.8 shows a virus of class V, in which the RNA genome serves instead as a *template* for mRNA synthesis. (This type is found in influenza viruses.) The RNA genome is transcribed into complementary RNA strands, which function both as mRNA and as templates for the synthesis of additional copies of genomic RNA. All viruses that use an RNA genome as a template for mRNA transcription require RNA → DNA synthesis. These viruses use a viral enzyme capable of carrying out this process; there are no such enzymes in most cells. The enzyme used in this process is encoded by the viral genome. After the protein is synthesized, it is packaged during viral self-assembly with the genome inside the viral capsid.

Table 19.1

Classes of Animal Viruses

A table titled Table 19.1 Classes of animal viruses illustrates class/ family, presence or absence of envelope, examples of viruses causing human diseases.

Class/Family	Envelope?	Examples that Cause Human Diseases

Roman numeral 1. Double-stranded D N A (d s D N A)		
Adenovirus (see figure 19.3b)	No	Respiratory viruses
Papillomavirus	No	Warts, cervical cancer
Polyomavirus	No	Tumors
Herpesvirus	Yes	Herpes simplex Roman numeral 1 and 2 (cold sores, genital sores); varicella zoster (shingles, chicken pox); Epstein-Barr virus (mononucleosis) Burkitt's lymphoma)
Poxvirus	Yes	Smallpox virus; cowpox virus
Roman numeral 2. Single-stranded D N A (s s D N A)		
Parvovirus	No	B 19 parvovirus (mild rash)
Roman numeral 3. Double-stranded R N A (d s R N A)		
Reovirus	No	Rotavirus (diarrhea); Colorado tick fever virus
Roman numeral 4. Single-stranded D N A (s s D N A; serves as m R N A		
Picornavirus	No	Rhinovirus (common cold); poliovirus; hepatitis A virus; other intestinal viruses
Coronavirus	Yes	Severe acute respiratory syndrome (S A R S); Middle east respiratory syndrome (M E R S)
Flavivirus	Yes	Zika virus (see figure 19.10c); yellow fever virus; dengue virus; West Nile virus; hepatitis C virus
Togavirus	Yes	Chikungunya virus (see figure 19.10b); rubella virus; equine encephalitis viruses
Roman numeral 5. s s R N A; serves as template for m R N A synthesis		

Filovirus	Yes	Ebola virus (haemorrhagic fever; see figure 19.10a)
Orthomyxovirus	Yes	Influenza virus (see figure 19.3c)
Paramyxovirus	Yes	Measles virus; mumps virus
Rhabdovirus	Yes	Rabies virus
Roman numeral 6. s s R N A; serves as template for D N A synthesis		
Retrovirus	Yes	Human immunodeficiency virus (H I V/ A I D S; see figure 19.9); human T- lymphotropic virus type-1 (H T L V-1) (leukemia)

The RNA animal viruses with the most complicated replicative cycles are the **retroviruses** (class VI). These viruses have an enzyme called **reverse transcriptase** that transcribes an RNA template into a DNA copy, an RNA → DNA information flow that is the opposite of the usual direction. This unusual phenomenon is the source of the name retroviruses (*retro* means “backward”). Of particular medical importance is **HIV(human immunodeficiency virus)**, the retrovirus shown in Figure 19.1 that causes **AIDS(acquired immunodeficiency syndrome)**. HIV and other retroviruses are enveloped viruses that contain two identical molecules of single-stranded RNA and two molecules of reverse transcriptase.

The HIV replicative cycle (illustrated in **Figure 19.9**) is typical of a retrovirus. After HIV enters a host cell, its reverse transcriptase molecules are released into the cytoplasm, where they catalyze synthesis of viral DNA. The newly made viral DNA then enters the cell’s nucleus and integrates into the DNA of a chromosome. The integrated viral DNA, called a **provirus**, never leaves the host’s genome, remaining a permanent resident of the cell. (Recall that a prophage, in contrast, leaves the host’s genome at the start of a lytic cycle.) The RNA polymerase of the host transcribes the proviral DNA into RNA molecules, which can function both as mRNA for the synthesis of viral proteins and as genomes for the new viruses that will be assembled and released from the cell. In Concept 43.4, we describe how HIV causes the deterioration of the immune system that occurs in AIDS.

Figure 19.9

The replicative cycle of HIV, the retrovirus that causes AIDS.

Detailed overview of the replicative cycle of H I V in 10 steps.

Replicative cycle of H I V.

1. The H I V envelope glycoproteins enable the virus to bind to specific receptors on certain white blood cells. Glycoproteins are shown on the outside of the viral envelope and the capsid is inside with two identical strands of R N A and reverse transcriptase.
2. The virus fuses with the cell's plasma membrane. The capsid proteins are removed, releasing the viral proteins and R N A.
3. Reverse transcriptase catalyzes the synthesis of a D N A strand complementary to the viral R N A.
4. Reverse transcriptase catalyzes the synthesis of a second D N A strand complementary to the first.
5. The double-stranded D N A is incorporated as a provirus into the cell's D N A.
6. Proviral genes are transcribed into R N A molecules, which serve as genomes for progeny viruses and as m R N As for translation into viral protein.
7. The viral proteins include capsid proteins and reverse transcriptase (made in the cytosol) and envelope glycoproteins (made in the ER).
8. Vesicles transport the glycoproteins to the cell's plasma membrane.
9. Capsids are assembled around viral genomes and reverse transcriptase molecules.
10. New viruses, with viral envelope glycoproteins, bud from the host cell. Side pictures show H I V entering a cell and leaving a cell (scale 0.25 micrometers).

The photos (artificially colored TEMs) show HIV entering and leaving a human white blood cell. See Figure 7.8 for the cell-surface proteins that act as receptors for HIV. Note in **STEP 6** that DNA synthesized from the viral RNA genome is integrated as a provirus into the host cell chromosomal DNA, a characteristic unique to retroviruses.

MAKE CONNECTIONS *Describe what is known about binding of HIV to immune system cells. (See*Figure 7.8*.)* How was this discovered?*

*For suggested answer, see*Appendix A*.*

HHMI Biointeractive Animation

HIV Replicative Cycle

image

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Animation

Retrovirus (HIV) Replicative Cycle

Evolution of Viruses

EVOLUTION We began this chapter by asking whether or not viruses are alive. Viruses do not really fit our definition of living organisms. An isolated virus is biologically inert, unable to replicate its genes or regenerate its own ATP. Yet it has a genetic program written in the universal language of life. Do we think of viruses as nature's most complex associations of molecules or as the simplest forms of life? Either way, we must bend our usual definitions. Although viruses cannot replicate or carry out metabolic activities independently, their use of the genetic code makes it hard to deny their evolutionary connection to the living world.

How did viruses originate? Viruses have been found that infect every form of life—not only bacteria, animals, and plants, but also archaea, fungi, and algae and other protists. Researchers estimate that each milliliter (one-fifth of a teaspoon) of seawater contains between one and 100 million viruses, ten times the number of microorganisms! Because viruses depend on cells for their own propagation, it seems likely that viruses are not the descendants of precellular forms of life but evolved—possibly multiple times—*after* the first cells appeared. Most molecular biologists favor the hypothesis that viruses originated from naked bits of cellular nucleic acids that moved from one cell to another, perhaps via injured cell surfaces. The evolution of genes coding for capsid proteins may have allowed viruses to bind cell membranes, thus facilitating the infection of uninjured cells.

Candidates for the original sources of viral genomes include plasmids and transposons. *Plasmids* are small, circular DNA molecules found in bacteria and in the unicellular fungi called yeasts. Plasmids exist apart from the genome, can replicate independently of the genome, and are occasionally transferred between cells. *Transposons* are DNA segments that can move from one location to another within a cell's genome. Thus, plasmids, transposons, and viruses all share an important feature: They are *mobile genetic elements*. (We'll discuss plasmids in more detail in Concepts 20.1 and 27.2 and transposons in Concept 21.4.)

Consistent with this notion of pieces of DNA shuttling from cell to cell is the observation that a viral genome can have more in common with the genome of its host than with the genomes of viruses that infect other hosts. Indeed, some viral genes are essentially identical to genes of the host.

Debate about the origin of viruses was reinvigorated about 20 years ago when an extremely large virus was discovered: Mimivirus is a double-stranded DNA (dsDNA) virus with an icosahedral capsid that is 400 nm in diameter, the size of a small bacterium. Its genome contains 1.2 million bases (Mb)—about 100 times as many as the influenza virus genome—and an estimated 1,000 genes. Perhaps the most surprising aspect of mimivirus, however, was that its genome included genes previously found only in cellular genomes. Some of

these genes code for proteins involved in translation, DNA repair, protein folding, and polysaccharide synthesis. Whether mimivirus evolved *before* the first cells and then developed an exploitative relationship with them or evolved more recently and simply scavenged genes from its hosts is not yet settled.

In the past decade, several even larger viruses have been discovered that cannot be classified with any existing known virus. One such virus is 1μm(1,000 nm) in diameter, with a dsDNA genome of around 2–2.5 Mb, larger than that of some small eukaryotes. What's more, over 90% of its 2,000 or so genes are unrelated to cellular genes, inspiring the name it was given, pandoravirus. The number of genes in pandoraviruses varies from 1,500 to 2,500 genes. A second virus, called *Pithovirus sibericum*, with a diameter of 1.5μm and 500 genes, was discovered in permanently frozen soil in Siberia. This virus, once thawed, was able to infect an amoeba after being frozen for 30,000 years! How these and all other viruses fit in the tree of life is an intriguing, unresolved question.

The ongoing evolutionary relationship between viruses and the genomes of their host cells is an association that continues to make viruses very useful experimental systems in molecular biology. Knowledge about viruses also allows many practical applications, since viruses have a tremendous impact on all organisms through their ability to cause disease.